

DEPARTMENT OF COMPUTER SCIENCE
ROLLWALA COMPUTER CENTER, GUJARAT UNIVERSITY
DEEP LEARNING
ASSIGNMENT-I - 2024

INSTRUCTIONS:

- You are expected to provide **detailed** responses to each question.
- Remember to make **honest** attempt.
- There are **4 sections**:
 1. Long answer type
 2. Logical thinking type
 3. Real life example or Application based type
 4. Complex and difficult type
- There are total **40 questions**, each section contains **10 questions**. Out Of These 4 Sections, Only **Long Answers And Logic Based** Question Will Be Covered For Exams.

Section I - Long-Answer Questions

1. Discuss the fundamental concepts of machine learning, focusing on supervised and unsupervised learning techniques. How do these concepts lay the foundation for understanding deep learning? Provide examples to illustrate your points.
2. Explain the structure and functioning of a deep neural network. What are the key differences between shallow networks and deep networks? Discuss how depth influences the learning capabilities of neural networks.
3. Define bias and variance in the context of machine learning models. Discuss the bias-variance tradeoff and its implications for model performance. How does this tradeoff manifest in deep networks, and what techniques can be employed to manage it?
4. Elaborate on the concept of the curse of dimensionality. How does it affect machine learning models, particularly deep networks? Discuss strategies that can be employed to mitigate the challenges posed by high-dimensional data.

5. Describe the architecture and functioning of a vanilla multilayer perceptron (MLP). How does an MLP differ from other types of neural networks, and what are its strengths and limitations? Include a discussion on activation functions used in MLPs.
6. Explain the role of flow graphs in deep learning. How does the backpropagation algorithm work within the framework of flow graphs? Discuss the mathematical principles behind backpropagation and its importance in training deep networks.
7. Discuss the Universal Approximation Theorem and its significance in the field of deep learning. How does this theorem justify the use of neural networks for a wide range of function approximations? Include an analysis of its limitations and practical implications.
8. Explain the concept of feature representation in deep learning. How does deep learning enable automated feature extraction, and why is this beneficial compared to traditional machine learning methods? Provide examples of how feature representation evolves across layers in a deep network.
9. Critically analyze how increasing the depth of a neural network impacts its learning capability and generalization performance. Discuss the challenges associated with training very deep networks, such as vanishing/exploding gradients, and the techniques used to address these issues.
10. How can deep learning models be designed and trained with bias-variance considerations in mind? Provide a detailed discussion on the methods for regularization, such as dropout and L2 regularization, and their role in controlling the complexity of deep networks.

Section II - Logical Thinking Questions

1. If a neural network with two hidden layers is overfitting the training data, what might be the effect of adding more hidden layers? Justify whether this change would likely improve or worsen generalization and why.
2. Given a model with high bias and low variance, what steps would you take to reduce bias while controlling for an increase in variance? Explain the trade-offs involved.
3. How would you expect the performance of a machine learning model to change if the dimensionality of the input data were doubled? What specific challenges might arise, and how could they be addressed?

4. If you are designing a neural network for binary classification, which activation function would you choose for the output layer and why? How does the choice of activation function impact the network's ability to learn?
5. Consider a deep network with several layers. If backpropagation were applied but with an incorrect flow graph (one that does not accurately represent the computational flow), what would be the likely result on the gradients? Discuss the consequences.
6. Given that the Universal Approximation Theorem states that a neural network can approximate any continuous function, why do some networks still perform poorly on certain tasks? What other factors must be considered beyond the theorem?
7. If a network is struggling to learn relevant features from input data, what adjustments could be made to the architecture or training process to improve feature representation? Discuss the role of layer depth and data augmentation.
8. How would the introduction of dropout in a deep network influence the model during training and inference? What logical reasoning underpins the effectiveness of dropout as a regularization technique?
9. If a deep network is showing signs of slow convergence or getting stuck during training, what logical explanation could be offered based on the vanishing/exploding gradient problem? What steps could be taken to address this issue?
10. Given two models: one with fewer parameters and another with significantly more parameters, which model would you expect to generalize better on unseen data, assuming both are trained on the same dataset? Provide logical reasoning for your answer, considering the concepts of bias, variance, and overfitting.

Section III - Application based questions

1. Medical Diagnosis Using Deep Networks

- How can feed-forward deep networks be applied to the early detection of diseases such as cancer or cardiovascular conditions? Discuss the challenges related to data quality, model interpretability, and the potential impact of bias and variance in medical diagnoses.

2. Autonomous Vehicles and Deep Learning

- Autonomous vehicles rely heavily on deep networks for tasks like object detection, lane keeping, and decision-making. Analyze the role of feed-forward deep networks in these systems,

particularly how they handle real-time data processing and the curse of dimensionality in high-resolution sensor data.

3. Deep Networks in Financial Fraud Detection

- Financial institutions use deep learning models to detect fraudulent activities in real-time. How can feed-forward deep networks be effectively designed to identify complex patterns of fraud while minimizing false positives? Consider the trade-offs in model complexity and real-time processing.

4. Deep Learning in Climate Modeling

- Climate models require the processing of vast amounts of multidimensional data to predict weather patterns and climate changes. Discuss how feed-forward deep networks can be applied to improve the accuracy of climate models. What challenges do these networks face with high-dimensional climate data?

5. Natural Language Processing for Customer Service

- Many companies use chatbots powered by deep networks to automate customer service. How can feed-forward deep networks be used to enhance the performance of these systems, particularly in understanding and generating natural language? Consider the balance between model complexity and response time.

6. Smart Cities and Urban Planning

- Urban planners use deep learning models to analyze satellite imagery and predict urban growth. How can feed-forward deep networks be employed to analyze patterns of urbanization and guide the development of smart cities? Discuss the role of feature representation in extracting meaningful insights from satellite data.

7. Personalized Recommendations in E-commerce

- E-commerce platforms like Amazon and Netflix use deep networks to personalize user recommendations. How can feed-forward deep networks be utilized to improve recommendation accuracy while addressing the curse of dimensionality due to large user and product feature spaces?

8. Deep Networks in Energy Demand Forecasting

- Energy companies rely on demand forecasting to optimize grid operations. How can feed-forward deep networks be applied to predict energy consumption patterns based on historical data? Discuss the challenges of feature selection and dealing with high-dimensional time-series data.

9. Agricultural Yield Prediction Using Deep Learning

- Predicting crop yields is critical for efficient agricultural management. How can feed-forward deep networks be used to model agricultural yield based on factors like weather, soil quality, and crop type? Analyze the potential impact of bias and variance in these models on prediction accuracy.

10. Deep Learning in Cybersecurity

- Cybersecurity systems often use deep networks to detect and respond to threats. How can feed-forward deep networks be applied to identify complex, evolving cyber threats in real-time? Discuss the trade-offs between model complexity, detection accuracy, and computational efficiency.

Section IV - Complex And Somewhat Challenging Questions

1. Analysis of Optimization Techniques in Deep Networks

- Discuss how the choice of optimization algorithm (e.g., SGD, Adam, RMSprop) affects the convergence behavior of deep networks. What challenges might arise when using these optimizers in extremely deep architectures, and how can they be mitigated?

2. Impact of Architectural Choices on Generalization

- How do various architectural choices, such as layer normalization, batch normalization, and skip connections, influence the generalization ability of deep neural networks? Evaluate the trade-offs involved in implementing these techniques in large-scale networks.

3. Theoretical Limits of the Universal Approximation Theorem

- While the Universal Approximation Theorem suggests that neural networks can approximate any continuous function, there are practical limitations. Analyze the implications of this theorem in the context of over-parameterization and the curse of dimensionality. How does this impact the design of network architectures?

4. Advanced Regularization Techniques

- Beyond traditional regularization methods like L2 regularization and dropout, discuss advanced techniques such as variational dropout and L1 regularization. How do these techniques balance the bias-variance tradeoff, and in what scenarios might they be preferable?

5. Exploring the Role of the Loss Function in Network Performance

- Evaluate how different loss functions (e.g., cross-entropy, hinge loss, mean squared error) influence the learning dynamics and final performance of deep networks. What considerations must be made when selecting a loss function for complex, multi-output tasks?

6. Curse of Dimensionality in Feature Spaces

- The curse of dimensionality is not just a concern in input data space but also in feature space. Analyze how deep networks attempt to mitigate the curse of dimensionality when mapping high-dimensional input to lower-dimensional representations. What role do techniques like PCA and t-SNE play in understanding this process?

7. Flow Graphs and Computational Efficiency

- Discuss the implications of different flow graph structures on the computational efficiency of training deep networks. How do choices in flow graph design impact the speed and memory usage during both the forward and backward passes? Provide a critical analysis of computational trade-offs.

8. Analyzing Deep Networks' Vulnerability to Adversarial Attacks

- Deep networks, particularly those used in image recognition, are vulnerable to adversarial attacks. Analyze how architectural decisions and training techniques influence a network's robustness against such attacks. What theoretical and practical measures can be implemented to enhance security?

9. Feature Extraction and Transfer Learning

- Transfer learning often leverages pre-trained deep networks as feature extractors for new tasks. Critically assess the conditions under which transfer learning is most effective. How does the choice of source domain and target domain affect the quality of the extracted features?

10. Interpreting the Black Box: Deep Network Explainability

- One of the criticisms of deep networks is their lack of interpretability. Discuss advanced methods such as SHAP values, LIME, and Grad-CAM for interpreting deep network decisions. Evaluate the effectiveness of these methods in providing transparency without compromising model performance. How do they align with the concept of feature representation?